

MI&CT ESERVICE NOTE
SN08021411a

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This short-term informational note is invalid 6 months after the issue date.

Date: February 14, 2008

Subject: Reset TNT Data Issues

Effectivity: Jedi HP-100/80/RT/VCT and Jedi HP-60/BrightSpeed

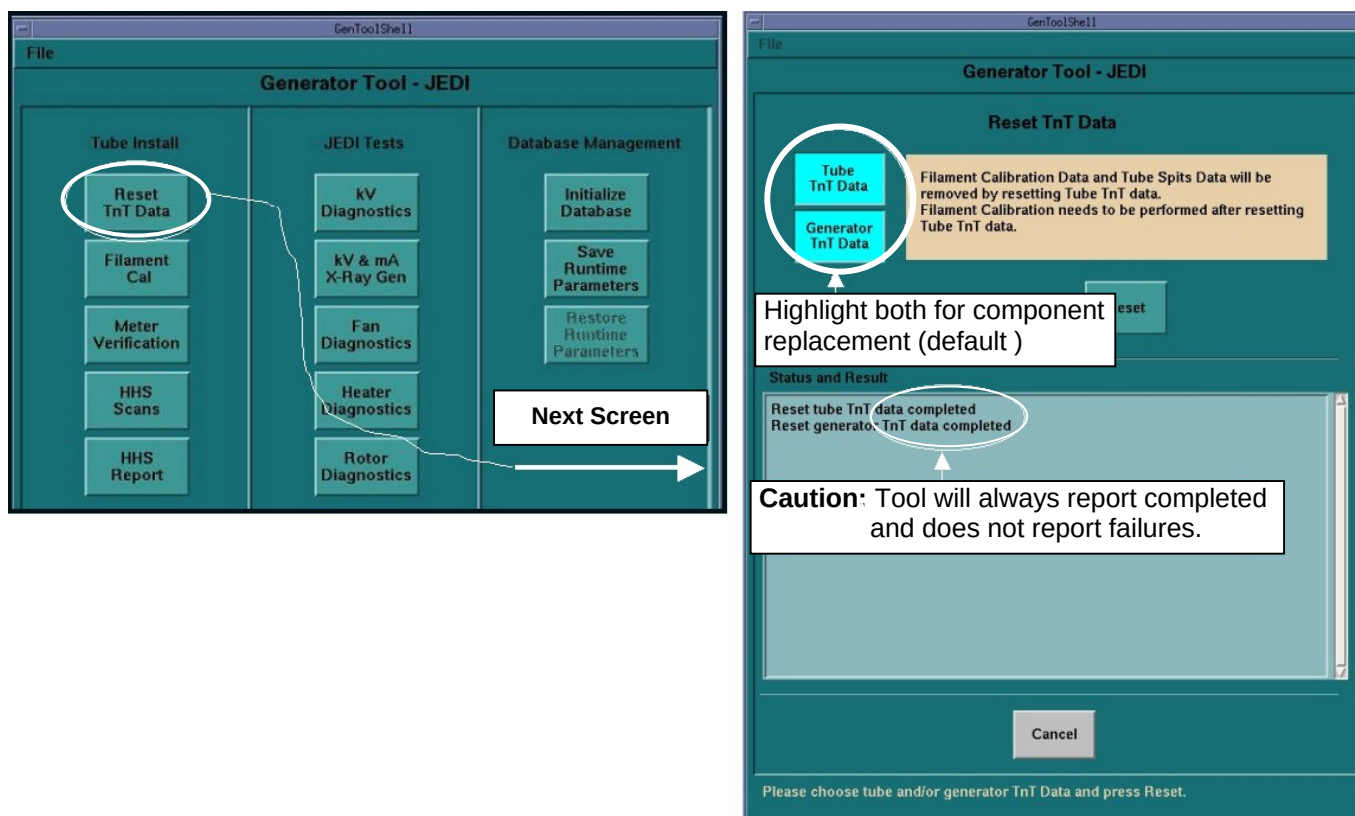
Details: The “Reset TnT Data” tool currently does not report failures in the GUI results window, the tool always reports “completed” whether it fails or succeeds. Successful completion of the “Reset TnT Data” tool is important during a tube change or generator component replacement because the tool resets the Generator and Tube TnT data (filament calibration, filament aging, statistics).

The “Reset TnT Data” tool is used to set the system back to a default filament calibration state therefore setting the mA/kV safe starting point. A successful TnT Reset will prevent damage to new components, e.g. tube, tanks. Basically “Reset TnT Data” tool sets a starting point for all large and small filament mA stations. Successful completion of Reset TnT Data tool is important during a tube change because an aged tube tends to have an increase in filament current to obtain the same tube output. This is called the filament aging adjustment. If a Reset TnT Data” is not performed on a new tube the new tube could show signs of spitting.

After the Reset TNT Data values are reset a complete Filament Calibration is Mandatory.

One common situation for “Reset TnT Data” failure can occur if the gantry “Emergency OFF” remains pressed during a hardware reset. This frequently happens after working in the gantry and forgetting to press “Reset” on the gantry controls after gantry 120VAC is turned ON.

Figure 1: Generator Tool – Reset TnT Data Screens



Resolution:

A successful reset of the TnT Data (starting filament current) is accomplished and verified by using the following steps.

- Before using Jedi generator tools such as Reset TnT, verify the gantry emergency stop is cleared by pressing the gantry controls "reset" button and then confirm a successful hardware reset.
 - Emergency Stop is cleared by pressing the gantry controls "reset" button
 - Perform a hardware reset.
- Procedure to verify a successful reset of the TnT data.** Verify the "Reset TnT Data" operation is successful by confirming the TnT Data is set to zero since the Reset TnT tool does not report failures in the GUI results window (as of March 2008).

Enter CSD – Diagnostics Menu

>Generator Tool –Jedi
> Reset TnT Data

- > Generator Tool - Jedi
 - > Kv & MA X-Ray Gen
 - > Upload TnT Data
 - > View TnT Data
- Refer to Figure 2; Verify the first line displays the current date and time, otherwise the data is from an older TnT Upload
 - If date/time is not current, repeat, "Upload TnT Data" and "View TnT Data"
 - If data/time is still not current, Jedi firmware is not responding. Check the gesyslog for failure causes.
- Verify all the values in the Tube TnT data are set to zero.**
 - If Tube TnT values are not zero as in figure 2 refer to the **Problem Area/Check Section** below.
- After a successful Reset of the TNT Data a complete Filament Calibration is Mandatory.

Figure 2: Generator Tool – kV/mA diagnostic - Upload/View TnT Data Screen

DIAGNOSTICS

Result

Tracking and Trending(TnT) Data: Tue Jan 8 09:10:34 2008

Tube TnT Data

Filament Aging Corrections for Small Filament:	0.000 A
Filament Aging Corrections for Large Filament:	0.000 A
Total Number of Spits:	0
Spits:	0
kV Regulation Error:	0
Number of Anode Accelerations:	0
Number of Exposure on Small Filament:	0
Number of Exposure on Large Filament:	0
Cumulated Energy on Small Filament:	0.000 kJ
Cumulated Energy on Large Filament:	0.000 kJ
Cumulated Rad Exposure Time on Small Focus:	0.000 S
Cumulated Rad Exposure Time on Large Focus:	0.000 S
Cumulated Fluoro Time:	0.000 S
Cumulated mAs:	0.000 mAs
Rotor PWM:	0.000 V DCBUS

Generator TnT Data

Max Temperature for IGBT:	0.00 deg
Max Temperature for Diode:	0.00 deg
Max Temperature for Snubber:	0.00 deg

Status

Start to update TnT data
Update TnT data has completed.

Problem Area/Check Section:

- I. Confirm "Tube TnT Data" button was selected (high-lighted) in screen 2 when "Reset" was pressed.
Note: "Generator TnT Data" is usually selected also but is not necessary to reset calibration data
- II. Check gesyslog for error messages relating to firmware DOWN, Jedi hardware, or gantry communication problems.
Example: gesyslog errors when a Jedi request fails.

```
Mon Jan 7 17:06:33 2008    Error: 244540
fwmgr_events.c          5614
Scan Seq Id: 0 Immediate Request cannot be honored since the firmware is in the DOWN state.
```

```
Mon Jan 7 17:06:33 2008    Error: 244634
Subsystem State; DCB:up CCB:up TCB:down JEDI:up TGP/ORP:up.
```

- III. Confirm gantry 120 VAC is ON, gantry controls Emergency OFF is reset, and hardware reset is successful.
Address any problems

Example: gesyslog message when Gantry Emergency OFF pressed (aka: Xray & drives OFF)

```
Fri Feb 1 14:10:05 2008      260112009      4
      tgp      tgp_subsys      uif_table_tilt_mgr_gt.c      1172
Host : Table Gantry Processor      Hermes # : 260112009
Exception Class : Operator Abort      Severity : Secondary
File : uif_table_tilt_mgr_gt.c      Line# : 1172
Function : SYSTEM POWER CONTROL : Emergency Stop
Scan Type : None/Unknown Scan: 0/0/0

Emergency Off button has been pressed
```

- VI. If High voltage disturbances are persistent utilizing the **"Initialize Database"** tool is recommended. Use the **"Save Runtime Parameters"** just prior to the selecting the **"Initialize Database"** preserves data for troubleshooting. **"Initialize Database"** tool replaces the entire Jedi database with a fresh copy containing blank TnT data. However this also clears the Jedi error log so if troubleshooting **"Save Runtime Parameters first"**. It is still a good practice to also verify TnT data is set to zero after using the **"Initialize Database"** tool, see page 2 for procedure to verify a successful reset of the TnT data.

Tracking Number (PQR/SPR/ECR): 13118873

CT Service

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